# Complex Numbers: Loci - Inequalities

Questions 1–15

**Question 1.** In each case, shade the region on the Argand diagram for the complex number $z$, labelling any points where loci intersect with the axes:

a) $|z| \geq 2$
b) $|z + 2i| < 4$
c) $|z - 3 + i| \leq 1$
d) $|z + 1 - i| > 5$
e) $2 \leq |z| \leq 4$
f) $|z - 1 - i| \geq 1$ and $|z - 2 - 2i| \leq 2$

**Question 2.** In each case, shade the region on the Argand diagram for the complex number $z$, labelling any points where loci intersect with the axes:

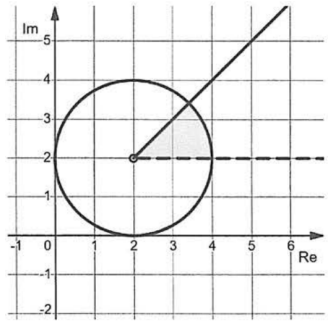
a) $0 < \arg(z - 2i) < \frac{\pi}{4}$
b) $\frac{\pi}{3} \leq \arg(z + 1) \leq \frac{2\pi}{3}$
c) $-\frac{\pi}{4} < \arg(z - 1 + 3i) \leq \frac{\pi}{3}$
d) $-\frac{\pi}{6} \leq \arg(z + 2 - i) < \frac{\pi}{6}$
e) $\arg(z - 4 + 2i) \leq \frac{2\pi}{3}$ and $\arg(z + 2 - i) \geq \frac{\pi}{4}$

**Question 3.** In each case, shade the region on the Argand diagram for the complex number $z$, labelling any points where loci intersect with the axes:

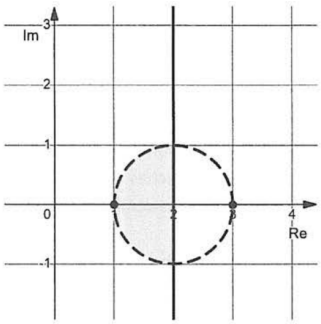
a) $|z-2| > |z-5|$
b) $|z-3| \geq |z+4i|$
c) $|z+1-2i| < |z-4i|$
d) $|z-4-3i| \leq |z+2-8i|$
e) $|z-1| > |z+3|$
and $|z-3-2i| \geq |z+1-4i|$

**Question 4.** For each of the following Argand diagrams state the conditions for the set of points $z$ represented by the shaded region. Give your answer in set notation.

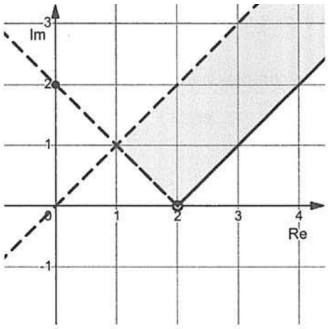
**a)**



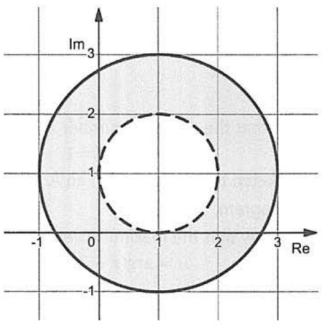

**b)**




**c)**




**d)**



**Question 5.**

a) On separate Argand diagrams shade the regions represented by $|z-3-4i| \leq 2$ and $0 \leq \arg(z-2-i) \leq \frac{\pi}{4}$

b) Hence shade the region satisfied by

$$\{z \in \mathbb{C} : |z-3-4i| \leq 2\}$$
$$\cap \left\{z \in \mathbb{C} : 0 \leq \arg(z-2-i) \leq \frac{\pi}{4}\right\}$$

**Question 6.**

a) On separate Argand diagrams shade the regions represented by $\left|z-2-\frac{1}{2}i\right| < 4$ and $\frac{\pi}{4} < \arg(z-3) \leq \frac{3\pi}{4}$. You do not need to label intersections with the axes.

b) Determine whether $2+\frac{1}{2}i$ lies in the region $\frac{\pi}{4} < \arg(z-3) \leq \frac{3\pi}{4}$

c) Hence, shade on an Argand diagram, the set of points

$$\left\{z \in \mathbb{C} : \left|z-2-\frac{1}{2}i\right| \leq 4\right\}$$
$$\cap \left\{z \in \mathbb{C} : \frac{\pi}{4} < \arg(z-3) \leq \frac{3\pi}{4}\right\}$$

**Question 7.**

a) Shade on an Argand diagram, the set of points

$$\{z \in \mathbb{C} : |z - 4 - 3i| \leq 2\}$$

b) Find the greatest and least value of $\arg z$ giving your answer in exact form.

**Question 8.**

a) Shade on an Argand diagram, the set of points

$$\{z \in \mathbb{C} : |z + 4 + 4i| \leq 2\}$$

b) Find the greatest and least value of $\arg z$ giving your answer to 3 significant figures.

**Question 9.** For the complex number $z$ find the maximum value of $|z|$ such that

a) $|z-4i| \leq 4$ and $0 \leq \arg z \leq \frac{\pi}{3}$

b) $|z-2| \leq 4$ and $\frac{\pi}{4} \leq \arg z \leq \frac{3\pi}{4}$

c) $|z-1-i| \leq 3$ and $|z-1-i| \geq |z+1-i|$

**Question 10.** For the complex number $z$ find the minimum value of $|z|$ such that

a) $|z-2-4i| \leq 2$ and $0 \leq \arg z \leq \frac{\pi}{4}$

b) $|z-3-3i| \leq \sqrt{2}$
and $\frac{\pi}{4} \leq \arg(z-1-i) \leq \frac{3\pi}{4}$

c) $|z-2| \leq 3$ and $|z+7-5i| \leq |z+3+7i|$

**Question 11.**

a) Shade on an Argand diagram, the set of points

$$\{z \in \mathbb{C} : |z - 2 - 2i| \leq 6\}$$

Label any intersects with the axes.

b) Determine whether $9\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$ lies in the region.

c) Determine whether $4\left(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right)$ lies in the region.

**Question 12.**

a) Shade on an Argand diagram, the set of points

$$\left\{z \in \mathbb{C} : \tan^{-1}\left(\frac{3}{4}\right) \leq \arg(z + 3i) \leq \frac{\pi}{2}\right\}$$

$$\cap\{z \in \mathbb{C} : |z - 1 + 2i| \leq 1\}$$

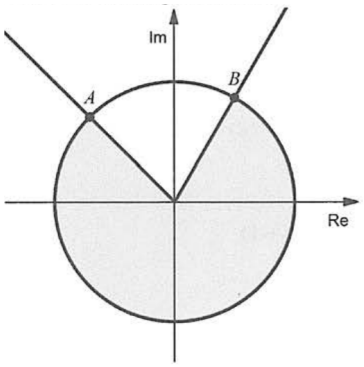
b) Find the value at which $|z|$ is a minimum.

**Prerequisite knowledge for Q13-Q15: Sector Area in Radians**

**Question 13.** A region may be defined by

$$\{z \in \mathbb{C} : |z| \leq 5\} \cap \left\{z \in \mathbb{C} : \frac{\pi}{3} \leq \arg z \leq \frac{3\pi}{4}\right\}$$

as shown in the diagram below:




a) Find the complex numbers representing points $A$ and $B$ on the diagram, giving your answer in exact form.
b) Find the area of the shaded region.

**Question 14.**

a) Shade on an Argand diagram, the set of points

$$\{z \in \mathbb{C} : |z - 4 - 3i| \leq 4\}$$

$$\cap\{z \in \mathbb{C} : |z - 2| \leq |z - 10|\}$$

b) Show that the area of the shaded region is

$$\frac{32}{3}\pi + 4\sqrt{3}$$

**Question 15.** Find the area of the region defined by

$$\{z \in \mathbb{C} : |z| \leq 2\} \cap \{z \in \mathbb{C} : |z - 3i| \leq 3\}$$

Give your answer to 3 significant figures.